HDOC DAY–7 ACTIVITY
C PROGRAMMING ASSIGNMENT REPORT

QUESTION 1 – Speed

Write a C program to find speed when distance and time are given.

1. Problem Understanding

- **Input:** Distance in meters (m) and time in seconds (s).
- **Output:** Speed in meters per second (m/s).
- **Formula:** Speed = Distance / Time.

Formula Summary

Quantity	Formula
Speed	v = d / t

2. Standard Test Cases

Test Case	Input (s)	Expected Output (s)	Actual Output (s)	Result (Pass/Fail)
1	d = 100 m; t = 5 s	Speed = 20.0000 m/s	Speed = 20.0000 m/s	Pass
2	d = 150 m; t = 10 s	Speed = 15.0000 m/s	Speed = 15.0000 m/s	Pass
3	d = 1000 m; t = 20 s	Speed = 50.0000 m/s	Speed = 50.0000 m/s	Pass

3. Algorithm

1. Start the program.
2. Declare variables for distance, time, and speed.
3. Read the distance in meters.
4. Read the time in seconds.
5. Calculate speed as distance divided by time.
6. Display the speed to four decimal places.
7. Stop the program.

4. C Program

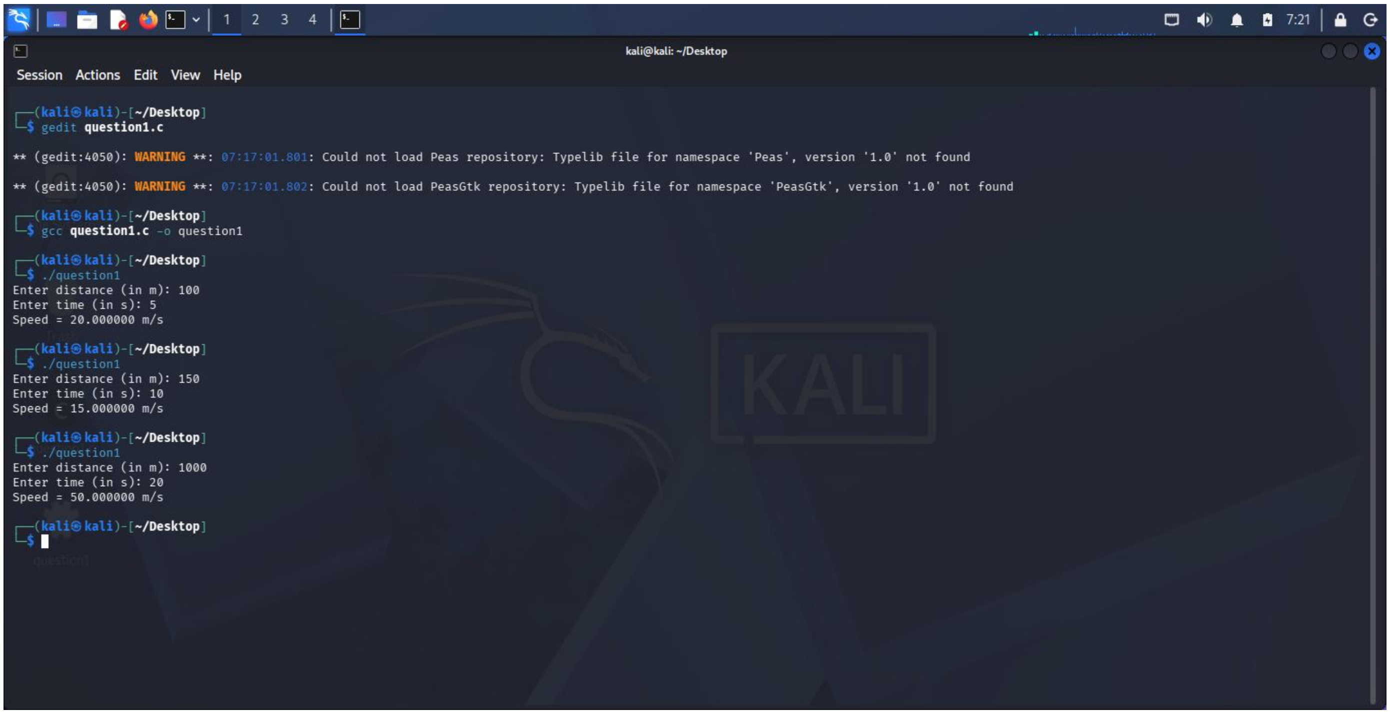
```
#include <stdio.h>
```

```
int main(void)

{

double distance, time, speed;

printf("Enter distance (in m): ");

scanf("%lf", &distance);

printf("Enter time (in s): ");

scanf("%lf", &time);

speed = distance / time;

printf("Speed = %f m/s\n", speed);

return 0;

}
```

5. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	d = 100 m; t = 5 s	Speed = 20.0000 m/s	Speed = 20.0000 m/s	Pass
2	d = 150 m; t = 10 s	Speed = 15.0000 m/s	Speed = 15.0000 m/s	Pass
3	d = 1000 m; t = 20 s	Speed = 50.0000 m/s	Speed = 50.0000 m/s	Pass

6. Compilation and Execution Evidence



QUESTION 2 – Force

Write a C program to find force when mass and acceleration are given.

1. Problem Understanding

- **Input:** Mass in kilograms (kg) and acceleration in meters per second squared (m/s^2).
- **Output:** Force in newtons (N).
- **Formula:** Force = Mass × Acceleration.

Formula Summary

Quantity	Formula
Force	$F = m \times a$

2. Standard Test Cases

Test Case	Input (s)	Expected Output (s)	Actual Output (s)	Result (Pass/Fail)
1	m = 10 kg; a = 2 m/s^2	Force = 20.0000 N	Force = 20.0000 N	Pass

2	m = 25 kg; a = 4 m/s^2	Force = 100.0000 N	Force = 100.0000 N	Pass
3	m = 50 kg; a = 9.8 m/s^2	Force = 490.0000 N	Force = 490.0000 N	Pass

3. Algorithm

1. Start the program.
2. Declare variables for mass, acceleration, and force.
3. Read the mass in kilograms.
4. Read the acceleration in meters per second squared.
5. Calculate force as mass multiplied by acceleration.
6. Display the force to four decimal places.
7. Stop the program.

4. C Program

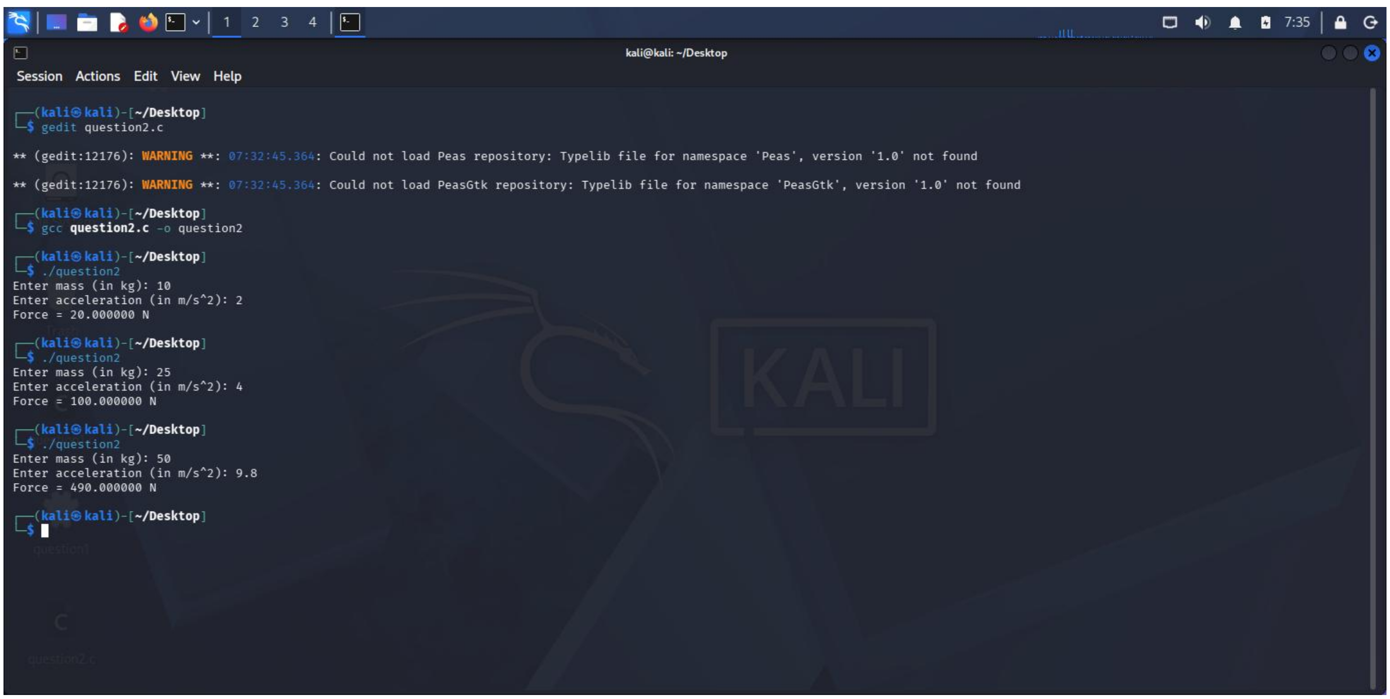
```
#include <stdio.h>

int main(void)

{

double mass, acceleration, force;

printf("Enter mass (in kg): ");

scanf("%lf", &mass);

printf("Enter acceleration (in m/s^2): ");

scanf("%lf", &acceleration);

force = mass * acceleration;

printf("Force = %f N\n", force);

return 0;

}
```

5. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)

1	m = 10 kg; a = 2 m/s^2	Force = 20.0000 N	Force = 20.0000 N	Pass
2	m = 25 kg; a = 4 m/s^2	Force = 100.0000 N	Force = 100.0000 N	Pass
3	m = 50 kg; a = 9.8 m/s^2	Force = 490.0000 N	Force = 490.0000 N	Pass

6. Compilation and Execution Evidence



QUESTION 3 – Electric Power

Write a C program to find electric power when voltage and current are given.

1. Problem Understanding

- **Input:** Voltage in volts (V) and current in amperes (A).
- **Output:** Electric power in watts (W).
- **Formula:** Electric Power = Voltage × Current.

Formula Summary

Quantity	Formula
Electric Power	P = V × I

2. Standard Test Cases

Test Case	Input (s)	Expected Output (s)	Actual Output (s)	Result (Pass/Fail)
1	V = 12 V; I = 2 A	Power = 24.0000 W	Power = 24.0000 W	Pass
2	V = 230 V; I = 5 A	Power = 1150.0000 W	Power = 1150.0000 W	Pass
3	V = 9 V; I = 1.5 A	Power = 13.5000 W	Power = 13.5000 W	Pass

3. Algorithm

1. Start the program.
2. Declare variables for voltage, current, and power.
3. Read the voltage in volts.
4. Read the current in amperes.
5. Calculate electric power as voltage multiplied by current.
6. Display the electric power to four decimal places.
7. Stop the program.

4. C Program

```
#include <stdio.h>

int main(void)

{

double voltage, current, power;

printf("Enter voltage (in V): ");

scanf("%lf", &voltage);

printf("Enter current (in A): ");

scanf("%lf", &current);

power = voltage * current;
```

printf("Electric Power = %f W\n", power);

return 0;

}

5. Verification of All Test Cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	V = 12 V; I = 2 A	Power = 24.0000 W	Power = 24.0000 W	Pass
2	V = 230 V; I = 5 A	Power = 1150.0000 W	Power = 1150.0000 W	Pass
3	V = 9 V; I = 1.5 A	Power = 13.5000 W	Power = 13.5000 W	Pass

6. Compilation and Execution Evidence